

```
This is a first test of sample test file
This is a second test of sample test file
```

Now, substitute the first occurrence of the pattern 'test' in each line with 'log':

```
$ sed 's/test/log/' sample.txt
```

```
This is a first log of sample test file
This is a second log of sample test file
```

If you want to substitute the second occurrence of pattern 'test' in each line with 'log', use the following commands:

```
$ sed 's/test/log/2' sample.txt
```

```
This is a first test of sample log file
This is a second test of sample log file
```

To substitute every occurrence of the pattern 'test' in each line with 'log', use the code below:

```
$ sed 's/test/log/g' sample.txt
```

```
This is a first log of sample log file
This is a second log of sample log file
```

The syntax for the above options is:

```
sed 's/original_pattern/replacement/options' <file_name>
```

Also, by default, *sed* will send the data to the standard output device, and you can redirect it to any file by using the redirection operator ">".

```
$ sed 's/test/log/g' sample.txt > mod_sample.txt
```

—Jagan Teki,
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Trace your route with *mtr*

mtr is a single network diagnostic tool that combines the functionality of the *traceroute* and *ping* programs.

mtr checks the network connection between the host it runs on, and *HOSTNAME*—by sending packets with deliberately low TTLs. It continues to send packets with low TTL, noting the response times of the intervening routers. This allows *mtr* to print the response percentage and response times of the Internet route to *HOSTNAME*. The output of the command indicates a bad or overloaded link if it shows packet loss or high response times.

The command can be used with different options to get the details of the host:

```
# mtr <option> <hostname/IP Address>
```

```
# mtr -c 3 -r linuxforu.com
```

The above command with options '-c 3 -r' will put *mtr* into report mode. When in this mode, *mtr* will run for the number of cycles specified by the -c option, and then exit after printing the statistics.

—Manikandan Samraj,
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Know details of the ports being used

Here is a simple trick to know whether or not a port on your Linux machine is being used by some process.

If you want to get information about port 80, type the following command at the terminal:

```
$ sudo fuser -n tcp 80
```

If you get an empty result, it means no one is using port 80.

If you get a result like what's shown below:

```
80/tcp: 1562 1651 1652 1653 1654 1655
```

...it shows the PID of the processes using port 80. Here, they are 1562, 1651, etc. It must show at least one PID.

Finally, take one PID (probably the first), and on the command line type the following:

```
$ cat /proc/1562/cmdline
```

You will get the name of the application using port 80. On my system, I got the result shown below:

```
/usr/sbin/apache2-kstart
```

So, you can try this magical procedure for any port.

—Tapas Saini,
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